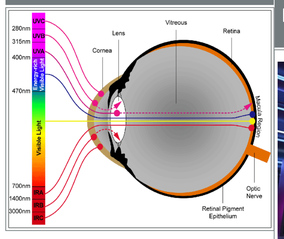
Heterogeneous Integration of Antimonide – Semiconductors with Si for Infrared Sensors

M. Muduli, Y. Xia, S. Lee, N. Gajowski, C. Chae, S. Rajan, J. Hwang, S. Arafin, and S. Krishna*

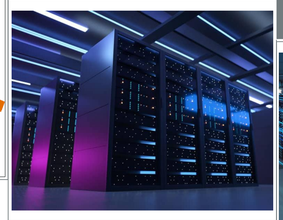
The Ohio State University
College of Engineering

MOTIVATION

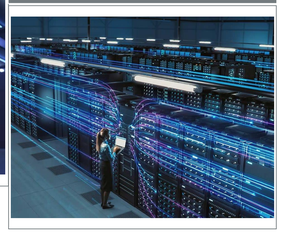
Eye-safe Wavelength:
1.55 µm Detector



Data Center



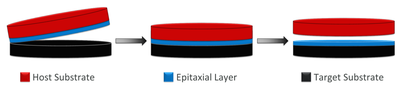
High-Speed Data Transfer



- No data center ≠ No internet fun
- High speed data transfer required between data centers
- Two main components: Lasers & Detectors

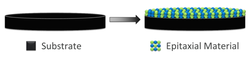
METHODOLOGY

1. Wafer Bonding

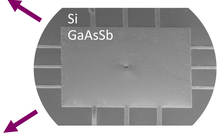


- Bonding of larger wafers
- Defects such as voids and cracks

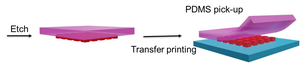
2. Epitaxial Growth



- Possibly high-quality material
- Challenging to grow III-V on Si

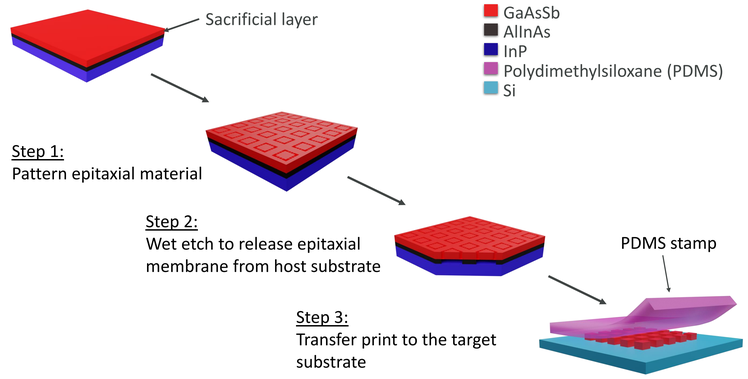


3. Epitaxial Layer Transfer



- Avoids interfacial defects
- Transfer nanomembrane or devices

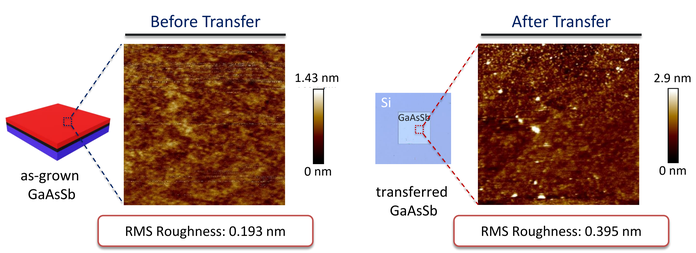
Epitaxial Layer Transfer



RESULTS & IMPLICATIONS

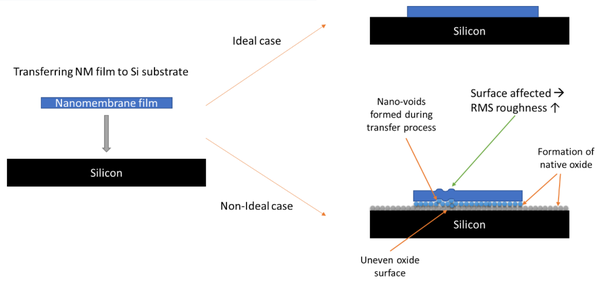
Surface Analysis

Atomic Force Microscopy



- RMS surface roughness increases after nanomembrane transfer

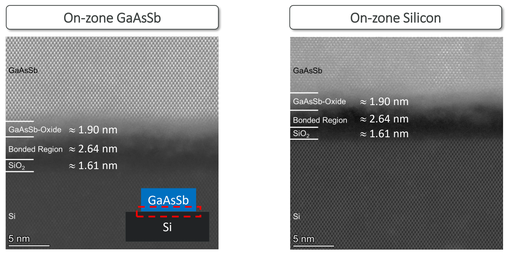
Why does the surface roughness increase?



- Surface roughness increases due to native oxide of the materials

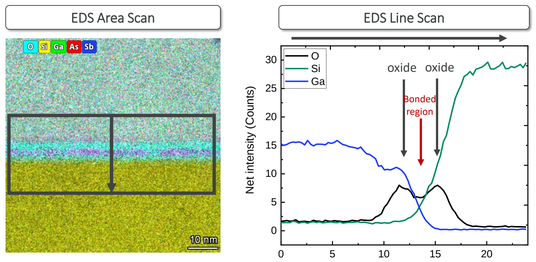
Interface Analysis

Transmission Electron Microscopy



- TEM data provides in-depth understanding of the interface of the detector
- This is important for understanding device physics and behavior

Energy Dispersive X-Ray Spectroscopy



- EDS scan provides elemental data of the interface & supports the TEM measurement

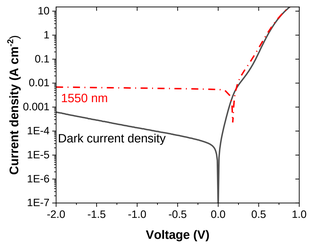
Device Characterization

Fabricated Devices



After transfer | After metal deposition | Laser illuminated device

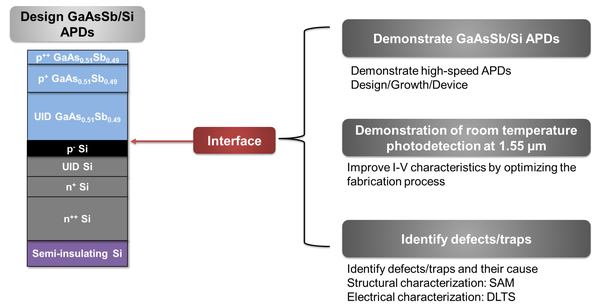
Current-Voltage Measurements



- Rectification behavior shows formation of a PN junction.
- Current-Voltage measurements show response to the eye-safe wavelength of 1.55 µm.

➔ Shows potential for high-speed detector
➔ Future application for high-speed data transfer

FUTURE WORK



REFERENCES

Scan the QR code to read the paper in detail!

[1] Muduli, M., Xia, Y., Lee, S., Gajowski, N., Chae, C., Rajan, S., Hwang, J., Arafin, S. and Krishna, S., 2024. Active interface characteristics of heterogeneously integrated GaAsSb/Si photodiodes. *Applied Physics Letters*, *125*(17).

[2] Abbasi, H.N., Lee, S., Jung, H., Gajowski, N., Lu, Y., Wang, Y., Kim, D., Zhou, J., Gong, J., Chae, C. and Hwang, J., 2024. Structural and electrical properties of grafted Si/GaAsSb heterojunction. *Applied Physics Letters*, *125*(10).

[3] Nallamothu, N.S., Xia, Y., Nikor, S.S.S., Jung, H., Gajowski, N., Lee, S., Arafin, S., Krishna, S. and Reano, R.M., 2023, October. Direct Bonding of GaAsSb to Silicon for High-Speed Avalanche Photodiodes. In *Laser Science* (pp. JM7A-105). Optica Publishing Group.

[4] Xia, Y., Nikor, S.S.S., Nallamothu, N.S., Adams, R.L., Jung, H., Gajowski, N., Lee, S., Reano, R.M., Krishna, S., Ringel, S. and Arafin, S., 2023, November. Fabrication of Si/GaAs 0.51 Sb 0.49 Heterostructure Diodes via Transfer Printing. In *2023 IEEE Photonics Conference (IPC)* (pp. 1-2).

KIND
Krishna INfrared Detector Laboratory

*Prof. Sanjay Krishna (Advisor)
George R Smith Professor of Engineering
Electrical and Computer Engineering

Abbreviations:-
- RT: Room Temperature
- Epi: Epitaxially grown layer
- AFM: Atomic Force Microscopy
- TEM: Transmission electron Microscopy
- EDS: Energy Dispersive X-Ray Spectroscopy
- SIMS: Secondary Ion Mass Spectroscopy
- XRD: X-Ray Diffraction
- APD: Avalanche Photodiode
- SAM: Scanning Acoustic Microscopy
- DLTS: Deep Level Transient Spectroscopy

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